

Algorithm for file updates in Python

Project description

I am a security professional working at a health care company. As part of your job, I am required to regularly update a file that identifies the employees who can access restricted content. The contents of the file are based on who is working with personal patient records. Employees are restricted access based on their IP address. There is an allow list for IP addresses permitted to sign into the restricted subnetwork. There's also a remove list that identifies which employees you must remove from this allow list.

My task is to create an algorithm that uses Python code to check whether the allow list contains any IP addresses identified on the remove list. If so, I should remove those IP addresses from the file containing the allow list.

Open the file that contains the allow list

```
# Assign 'import_file' to the name of the file
import_file = "allow_list.txt"
```

```
# Build with 'statement' to read in the initial contents of the file
with open(import_file, "r") as file:
```

Read the file contents

```
with open(import_file, "r") as file:
# Use 'read()' to read the imported file and store it in a variable named "ip_addresses"
ip_addresses = file.read()
```

Convert the string into a list

```
# Use 'split()' to convert 'ip_addresses'
from a string to a list
ip_addresses = ip_addresses.split()
```

Iterate through the remove list

```
# Build iterative statement
# Name loop variable 'element'
```

```
# Loop through  
'remove_list'  
for element in remove_list:
```

Remove IP addresses that are on the remove list

```
for element in remove_list:  
# Create conditional statement to evaluate if 'element' is in 'ip_addresses'  
if element in ip_addresses:
```

```
# use the 'remove()' method to remove  
# elements from 'ip_addresses'  
ip_addresses.remove(element)
```

Update the file with the revised list of IP addresses

```
# Convert 'ip_addresses' back to a string so that it can be written into the text file  
ip_addresses = "\n".join(ip_addresses)
```

```
#Build 'with' statement to rewrite the original file  
with open(import_file, "w") as file:
```

```
# Rewrite the file, replacing its contents with 'ip_addresses'  
file.write(ip_addresses)
```

Summary

I created an algorithm that removes IP addresses listed in a `remove_list` variable from the "allow_list.txt" file, which contains approved IP addresses. The process starts by opening the file, reading its contents as a string, and converting that string into a list stored in the variable `ip_addresses`. Then, I looped through each address in the `remove_list` and checked if it appeared in `ip_addresses`. If it did, I used the `.remove()` method to delete it from the list. Finally, I used the `.join()` method to turn the list back into a string and rewrote the "allow_list.txt" file with the updated IP addresses.